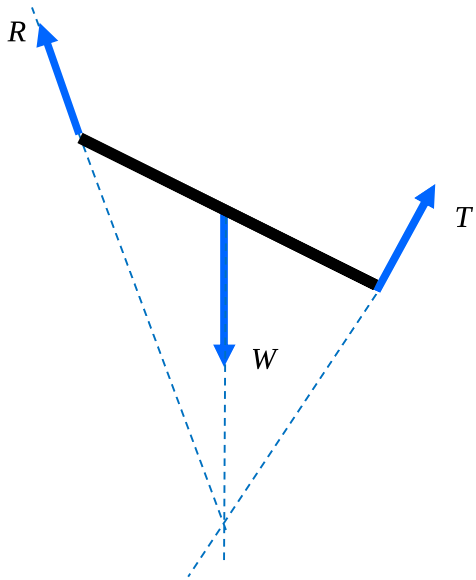
Examiner's Report and Solution for 2020 SAJC H2 Physics Prelim Paper 2

1 (a)	<u>Resultant/ net force</u> is zero <u>Resultant/ net moment</u> about <u>any axis/point</u> is zero	[1] [1]
(b)(i)	 • T and W correctly labelled with full name, T and W in correct direction. • All 3 forces intersect to meet at a common point (clearly shown with dotted lines!) and R correctly labelled & in correct direction *W* is the weight of the rod, *T* is the tension acting on the rod, *R* is the force acting on the rod by the hinge/ contact force	[1] [1]
(b)(ii)	Take moments about the hinge, $30 \times 9.81 \times \cos 30^o (\frac{L}{2}) = T \times L$ $T = 127.4 \approx 127 N$	[1, 1]
(b)(iii)	$(\leftarrow +) \quad R_X = T \sin 30^o = 127.4 \sin 30^o = 63.7 N$ $(\uparrow +) \quad R_Y = 30 \times 9.81 - T \cos 30^o = 184 N$ $R = \sqrt{R_X^2 + R_Y^2} = \sqrt{63.7^2 + 184^2} = 194.7 \approx 195 N$ ecf allow $\tan\theta = \frac{184}{63.7} \Rightarrow \quad \theta = 70.9^o$ <u>above the horizontal</u> as shown in Fig. 1 ecf allow	[1] [1] [1] [1]